



Customer Shopping Behavior Analysis

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.



Dataset Summary: Unpacking the Data

Rows & Columns

3,900 rows and 18 columns provide a rich dataset for analysis.

Key Features

Includes demographics, purchase details, and shopping behavior metrics.

Missing Data

37 values in the 'Review Rating' column required imputation.

Exploratory Data Analysis (EDA) with Python

Our initial phase involved comprehensive data preparation and cleaning using Python, ensuring data quality and readiness for in-depth analysis.

01

Data Loading & Exploration

Imported dataset with pandas; used `df.info()` and `.describe()` for initial insights.

02

Missing Data Handling

Imputed missing 'Review Rating' values using the median rating per product category.

03

Column Standardization

Renamed columns to 'snake_case' for improved readability and consistency.

04

Feature Engineering

Created 'age_group' and 'purchase_frequency_days' for deeper segmentation.

05

Database Integration

Loaded the cleaned DataFrame into PostgreSQL for subsequent SQL analysis.

SQL Analysis: Key Business Insights

We leveraged PostgreSQL to conduct structured analysis, addressing critical business questions about customer behavior and sales performance.

Revenue by Gender



Male customers generated significantly more revenue (\$157,890) than female customers (\$75,191).

Top 5 Products by Rating



Gloves, Sandals, Boots, Hat, and Skirt were the highest-rated items, indicating strong customer satisfaction.

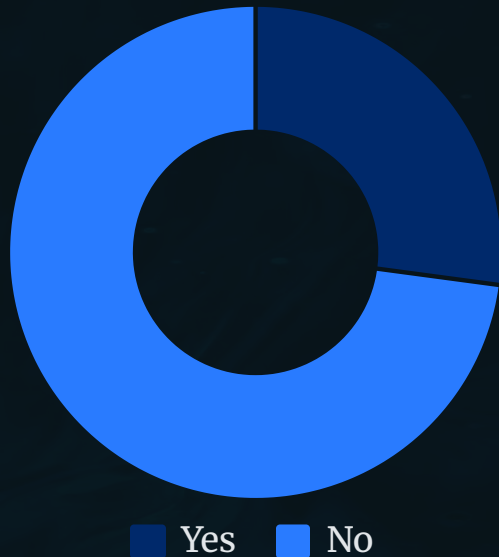
Shipping Type Comparison



Express shipping users had a higher average purchase amount (\$60.48) compared to standard shipping (\$58.46).

Subscription Status & Spending

An analysis of subscription status reveals interesting patterns in customer engagement and spending habits.



While subscribers represent a smaller portion of the customer base, their average spend is comparable to non-subscribers.

Yes	1053	59.49	62645.00
No	2847	59.87	170436.00

Non-subscribers contribute significantly more to total revenue due to their larger numbers.



Discount Impact & Product Performance

Understanding how discounts influence purchases and identifying top-performing products are crucial for optimizing sales strategies.



High-Spending Discount Users

839 customers used discounts but still spent above the average purchase amount, indicating effective discount targeting.

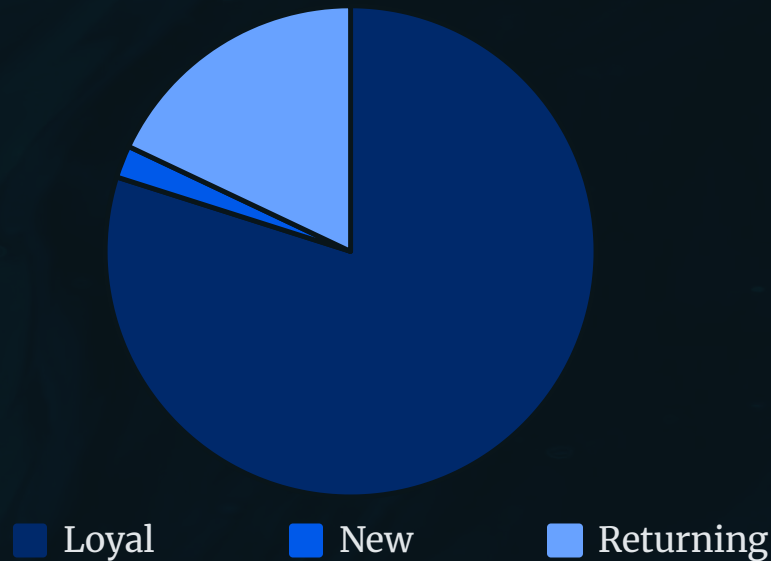


Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%) had the highest discount rates.

Customer Segmentation & Loyalty

Segmenting customers by their purchase history allows for tailored marketing efforts and loyalty programs.



The majority of customers fall into the 'Loyal' segment, highlighting a strong existing customer base.

1

Loyal Customers

3,116 customers demonstrate consistent purchasing behavior.

2

Returning Customers

701 customers have made multiple purchases, but not yet reached loyal status.

3

New Customers

83 new customers represent growth opportunities.

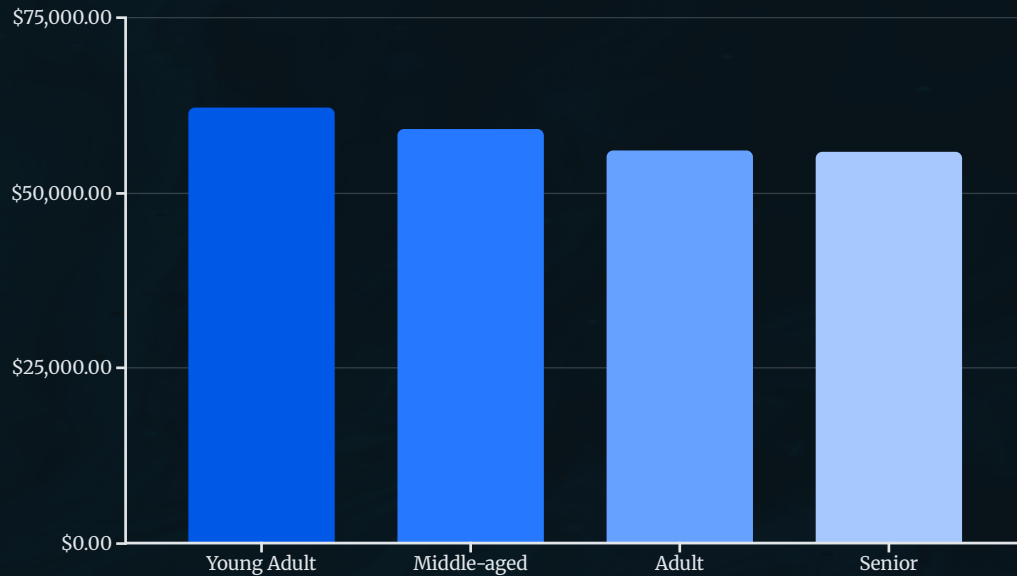
Top Products by Category

Identifying the most purchased items within each category helps in inventory management and targeted promotions.

Accessories	Jewelry	171
Accessories	Sunglasses	161
Clothing	Blouse	171
Clothing	Pants	171
Footwear	Sandals	160
Footwear	Shoes	150
Outerwear	Jacket	163
Outerwear	Coat	161

Revenue by Age Group & Repeat Buyers

Understanding the revenue contribution of different age groups and the subscription likelihood of repeat buyers informs targeted marketing and retention strategies.



Young Adults contribute the highest revenue, followed closely by Middle-aged customers.

Repeat Buyers

Customers with more than 5 purchases are more likely to subscribe (958 subscribers vs. 2518 non-subscribers).



Subscription Potential

Targeting repeat buyers for subscription offers could significantly boost subscriber numbers.

Business Recommendations: Strategic Growth

Based on our comprehensive analysis, we propose actionable strategies to enhance customer engagement, optimize sales, and drive sustainable growth.

1

Boost Subscriptions

Promote exclusive benefits to increase subscriber base and recurring revenue.

2

Customer Loyalty Programs

Implement reward systems to foster loyalty and encourage repeat purchases, moving customers into the "Loyal" segment.

3

Review Discount Policy

Strategically adjust discount offerings to balance sales boosts with healthy profit margins.

4

Product Positioning

Highlight top-rated and best-selling products in marketing campaigns to maximize visibility and sales.

5

Targeted Marketing

Focus marketing efforts on high-revenue age groups and users who prefer express shipping for optimized ROI.